

Do humans process (also) qualitative information? Help from computational metaphysics welcome!

Martin Korth, U Münster

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From artificial to human intelligence

- ▶ **'sub-symbolic' deep neural network (DNN) based artificial intelligence (AI)** runs into all sorts of problems – due to missing world models? (vs superposition of all world models compatible with statistics of language token use in training data?)
- ▶ **classical, symbolic AI** systems run into problems with the generation and operation of open, extended world models
- ▶ **combined ('Neuro-AI' or similar) approaches** additionally run into problems with interfacing the two in a general way
- ▶ so how do humans generate or access (layers of) symbol systems and world models?
– **we don't know**
- ▶ this points to core philosophical questions: what is **abstraction**? are there proper (Platonic) **abstract objects**? etc.
- ▶ AI research and development offers one of many different routes towards the insight that **something about abstract objects is very 'real' ...**
- ▶ ... a lot of new problems are old problems ...

Related 'old' problems

- ▶ the '**symbol grounding problem**' of the connection between data and meaning
- ▶ the '**framing problem**' of implementing common sense knowledge and know-how
- ▶ the '**hard problem**' of phenomenological experience
- ▶ the problem of (broad and stable) **abstraction**
- ▶ the problem of (also the implementation of) **universals**

→ in all these cases, 'meaning' seems to evade a purely relational definition, and seems to point to the existence of qualities, forms, universals, etc.

Quantitative vs qualitative information

- ▶ a helpful distinction from the Philosophy of Information:
 - ▶ **quantitative information**; physically implemented, relationally defined
 - ▶ **qualitative information**; 'meaning' in the (original) sense of Frege's 'Sinn'
- ▶ hitherto still unclear how exactly the two are related
- ▶ **both sub-symbolic, as well as symbolic AI run on quantitative information**, though on different levels; the problem of interfacing them is connected to the (arbitrarily) different relational assignment of 'functional' meaning to the respective symbols

Quantitative vs qualitative information -2-

- ▶ **human brains** seem to work to a good extent like DNNs, with functionally understood 'meaning' as quantitatively optimized, relational structure, i.e. quantitative information
- ▶ at some point above(?) this, **human minds** seem to work with bits of meaning (most basic: qualia), i.e. qualitative information
- ▶ Gödel: 'The basis of everything is meaningful predication ... Husserl had this, Hegel did not have this: that is why his philosophy lacks clarity ...'
- ▶ **most straightforward way to conceptualize qualitative information is via distinct properties – then giving rise to abstract entities** (the question of pre-existence of abstract objects is secondary to this)

Two common ways to approach the problem

(some sort of) Humean Supervenience (Hume, Lewis, ...)

- ▶ **abstract objects as regular patterns found in concrete objects**
- ▶ connects to current AI research, common theories of consciousness, connectionist neuroscience, materialist frameworks, etc → **deflated view of human thought**
- ▶ but people are objects themselves; **how exactly is human thinking implemented?** (the above definition is clear for logic, but not so much for neuroscience)
- ▶ this does not seem to work out (yet?); **we run into 'old' problems**

Platonic abstract objects (Plato, Leibniz, Peirce ...)

- ▶ **abstract objects (at least) as real as concrete ones**
- ▶ bundles of universal, qualitative properties; carrying all sorts of 'meaning'
- ▶ not relationally defined (via web of words, possible worlds etc.), but vice versa
- ▶ **(higher-level) human thought then manipulates actual (though abstract) objects**
- ▶ our 'world model' is then some form of complex, brain-anchored abstract object and requires a complex sub-consciousness – **can this work out scientifically?**

Would Gödel have voted for the later option? – We actually want both!

In the language of computational metaphysics ...

- ▶ how to **differentiate** *and interface* **abstract and concrete objects**
- ▶ possible from the **logical side** via (a modernized understanding of) Mally's distinction of **exemplification vs encoding** and then modern **object theory** (Zalta *et al.*)
- ▶ less clear how to proceed from the **scientific side**; how do concrete objects process abstract ones, i.e. **how to scientifically bridge this (effectively dualist) distinction?**

- ▶ if encoding is realized as **physical information processing**, how does this not run into the '**old**' **problems?** (qualia seem to be exemplifications, not encodings, etc.)
- ▶ if encoding is related to (proper) **Platonic abstract objects**, then exemplification is different only due to **causal couplings that are in need of scientific explanation** (closer to Meinong than Mally?)

→ back to the question: **can the later work out scientifically?**

Model A or 'A-world'

- ▶ specific proposal for how to solve the **emanation problem**, i.e. how to recover modern science **in detail**, within models like the ones of Plato, Leibniz, Gödel ...
- ▶ illustrates more generally which problems have to be solved; puts all pressure on science (which is good at re-inventing ontology!) to make this a 'workable' project
- ▶ **core problem is thus the micro-structure of our world** – here we have to make specific ontological commitments; what is matter? wherefrom causation? etc.
- ▶ interaction & constitution/combination problems lead to monadic idealism
→ '**Plato & Leibniz updated for modern science**' – platonic & scientific realism
- ▶ **world fundamentally build** not from particles or fields in spacetime, but more generally **from universal, qualitative properties**
- ▶ defense: show compatibility down to mathematics of our best scientific theories
- ▶ offense: think through the consequences to devise **new experiments**
→ a scientific approach to (then pragmatic) idealism ('reverse Gödel')

→ see M. Korth, *Information, Intelligenz and Idealismus*
Brill/Mentis 2026, English manuscript on PhilPapers.org

New experiments?!? – (some) conceivable consequences

Effects from the universal nature of the building blocks

- ▶ quantum nature of reality at micro-scale
- ▶ relativity as necessity for identity preservation
- ▶ ...

Possible advantages from handling universals

- ▶ stable biological organization via non-physical feedback markers?
- ▶ **information processing cut short?**
- ▶ ...

Processes/structures across the Neuroscience/Psychology gap

- ▶ physical memory partly as a register for non-physical structures?
- ▶ 'super-coordination' of neural activity? (still with a lot of constraints!)
- ▶ ...

The example here: Information processing cut short?

If humans process qualitative information

- ▶ then this should be connected to **evolutionary advantages** over purely quantitative information use, and some advantage would need to have been **accessible early-on**
 - **what is the 'evolutionary purpose' of qualia?** (= one-property abstract objects)

What's special about qualitative information?

- ▶ this question is itself part of the project to tackle the problem
- ▶ **working hypothesis:** qualitative information is 'transparently' given, not relationally defined; and 'works' via some form of material inference or 'semantic logic/aesthetic'
 - **most straight-forward way to conceptualize this is Platonic realism** → whatever is special about abstract objects is special about qualitative information
- ▶ **connects to object theory in (albeit here stronger) Platonic interpretation**
 - see E. Zalta's contribution to 2025 Kurt Gödel Essay Prize

An example hypothesis ...

The ability to process qualitative information allows humans to avoid the problems of classical AI (and combined approaches) with generating and handling complex world models

- ▶ **universal** → world models automatically contain each abstract entity only once; instances are connected to 'classes', existing relations are automatically imported
- ▶ **qualitative** → all conceivable negations are automatically implemented
- ▶ **abstract** (= w.r.t. material consistency requirements incomplete) → incomplete entities behave 'quantum object-like' within world models:

'Your cat' collapses from 'superposition' of all conceivable cats to a specific cat once I encounter it (the abstract/concrete distinction is orthogonal to this!)

- ▶ Wait, isn't this just what abstract objects do, Platonic or not?
 - Yes, but here without additional physical computation efforts for 'binding' etc.
- **we should be able to distinguish the two cases w.r.t. brain structure & activity**

... and an example (so far thought-)experiment

Compare computational effort for object recognition

- ▶ as a first step in 'binding'; i.e. placing properties/objects in world model
- ▶ **(naive) symbolic AI**: list search scales $\propto$ number of properties n and number of identifiable objects N , overall $O(n \cdot N)$
- ▶ **(naive) sub-symbolic AI**: object recognition scales $\approx O(N)$
- ▶ **quantum computing via Grover algorithm**: list search scales $\propto n$ and the number of amplification steps to reach the required accuracy, overall $O(n \cdot \sqrt{N})$
- ▶ **universal qualitative property handling**: object 'setup' depends only on the number of properties (which negate all alternatives), i.e. overall $O(n)$ – like Grover w/o amplification – and automatically imports all background knowledge?

BUT how to investigate this separately from the underlying quantitative processing?
(which would have to initiate the delivery of qualia to our subconsciousness)

AND which specific algorithms do we use for comparison in experiments?

→ we need models like A-world to map out hypothetical details to test for;
the search for consequences should then be formalizable in object theory

Thank you for your attention!

Take home message:

If Platonic abstract objects exist and if humans can process them, then we should be able to find experimental evidence for this at the Neuroscience/Psychology interface

more: M. Korth, *Information, Intelligenz und Idealismus*
Brill/Mentis 2026, english manuscript on PhilPapers.org:



Appendix I:

Ontology, Gödel, perception vs imagination, private qualia, identity & 'anchoring', neuroscience

Does the ontology of such models include an infinity of basic properties?

In Model A

- ▶ this is first of all an empirical question similar to how we do not know about physics before experiments
- ▶ seemingly infinite 'ranges' of properties (color shades etc.) might be the result of the bundling of simple ones
- ▶ Platonic realists (and many scientists implicitly) usually accept an infinity of at least basic mathematical entities

Do all abstract objects pre-exist in such models?

In Model A

- ▶ non-physical, basic properties, but not complex abstract entities are pre-existing
- ▶ complex entities form in individualized minds based on physical signals being converted to non-physical properties and then simple abstract objects by our sub-consciousness
- ▶ against the backdrop of our existing world model, physical signals would have to be helpful enough to enable this
- ▶ shared biology would guarantee (at least partially) successful communication
- ▶ universality of the basic building blocks leads to seemingly universal nature of complex abstract objects

A Gödelian outlook

For models like A-world we find ...

- ▶ ... conceptual and mathematical realism
- ▶ ... physical (unlike subjective) time to be not fundamental, but the outcome of the development of a material part of reality, i.e. of a subset of objects with properties which change according to consistency rules
- ▶ ... ontological arguments to be 'trivially' correct, but without immediate necessary consequences for the material world, as they prove abstract existence only

The difference

Gödel – rational optimism

- ▶ the pre-supposed (wished-for?) effectiveness of logic to deal with ontology requires necessary implications between properties ('epistemic stability' → see L. Hornischer's contribution to 2025 Kurt Gödel Essay Prize)
- ▶ this leads to all sorts of 'eternal truths' (Hegel's semantic logic made proper, so to say); we can then identify positive properties, that existence is positive, etc.
- ▶ all contingency, including its yardstick physical time, ends up to be just down to our limited experience of a perfectly well-ordered whole, implying theism

A-world – pragmatic idealism

- ▶ idealism not as rationally derivable outcome, but as something scientific inquiry might establish as best overall fit
- ▶ allows for contingent understanding of positive properties, existence, ...
- ▶ ontological arguments then seem to imply at the meta-level that if there is a strict 'semantic logic' (instead of a 'semantic aesthetic') between properties, then all abstract entities are (pre-)existing – which would take us back to Plato's Meno ...
- ▶ rational optimism would then be justified

How would imagination and perception differ?

In Model A

- ▶ first of all, this is again an empirical question
- ▶ the **boundary is not as rigid as we often think** (drugs, schizophrenia, meditation)
- ▶ a straightforward hypothesis would be that the two happen at **different levels of abstraction**
- ▶ with **perception/qualia** as 'pushed' by our consciousness ...
- ▶ ... and **imagination** most often happening at a level where more easily manageable concepts stand in for the complex objects of perception

- ▶ the necessity of such a distinction between qualia and concepts of qualia was anyhow proposed in Philosophy by M. Nida-Rümelin
- ▶ **'proper encoding'/abstraction** in the common sense at the higher level(s)
- ▶ the **activation of perception in imagination** would remain a possibility, e.g. in those special cases mentioned above

but why/how would such a layered mental structure have evolved?

why/how would such a layered mental structure have evolved?

- ▶ the **direct handling of perceptual objects** would always take them and their implications as true; this would work like what we expect for **instincts**
- ▶ 'Zoos are not for animals' - certain reactions cannot be avoided
- ▶ being able to push qualitative signals to a **more abstract level would take pressure from lower levels to immediately follow implications**
- ▶ this allows for reflection and (attempts at) global optimization
- ▶ once such a mechanism is available, higher-level abstraction is in reach

- ▶ one powerful feature of language would then be to indicate 'here is a thing', without attaching any (or more than the minimum?) of properties to it
- ▶ (large parts of?) abstract thought would then indeed be conceptual, but taking all of human thought as language-based would still not be right
- ▶ the hypothesis above might be experimentally testable: do ('lower') animals remember individuals? do we not remember individuals in detail? etc.

Are qualia not privately generated and exclusive?

In Model A

- ▶ physical and non-physical properties on the same footing:
 - ▶ individual material objects from 'particle bundles' of universal material properties, e.g. universal spin in concrete particle
 - ▶ individual non-material objects from bundles of universal non-material properties, e.g. universal red in concrete mental content
- ▶ universal material properties essential to our understanding of laws of nature
- ▶ are universal non-material properties the basis for successful intersubjective communication beyond 'success in material world' cases?
- ▶ **very hard to conceive science-like models of phenomenological experience without universal non-material properties**
- ▶ would correspond to each particle having its own spin-property, so that we would have to check for every particle pair according to which pair-specific law their individual spin-properties would interact
- ▶ we are happy to embrace Platonism for physical properties (universal mass, spin, etc. properties for all particles) – should be our first guess for mental properties, too

How would our minds be anchored to ensure their identity in a world of universals?

Starting from universals and the evolution of a monad population only ...

- ▶ **easy loss of identity at micro-scale**, because any conceivable particles are abstract objects as long as their properties are not handled consistently
- ▶ spatial properties are one way out; **space for identity, not identity from space**
- ▶ stable identities can form at the molecular scale thanks to **evolutionary learned consistency rules** for handling the subset of properties that make up the physical world
- ▶ when monads reach out to non-physical properties, these must be 'anchored' in individualized bodies for stable growth (scepticism and truth become issues then)
- ▶ only complex minds do then allow for the 'free' handling of abstract objects (anchored in a world model, anchored in a body, etc.)
- ▶ and these minds have to have a complex (and multi-monadic) sub-consciousness to handle both physical and non-physical properties by different rules
- ▶ would lead to **additional 'neural design principles'** (besides energy optimization) like 'pass everything important through matter'
- ▶ **alternatives:** pre-established harmony, dispositional properties, ...

Neuroscience & Model A

Neural processing of visual information

- ▶ **structures known:** photoreception, bipolar & ganglion cells, LGN, V1-V5 areas, ...
- ▶ **pathways known:** dorsal stream or 'where/how pathway' (semi-automatic, vision-for-action) for location, motion, spatial relationships; ventral stream or 'what pathway' (executive control, vision-for-perception) for object details
- ▶ higher up involvement of areas for attention, memory, decision making, ...
- ▶ prefrontal cortex (memory, decision-making) for maintaining & manipulating objects (decision-making as critical component of object recognition)
- ▶ temporal lobes (long term memory, semantic processing) for contextual information
- ▶ **we can pin down brain areas & pathways that are sensitive to certain object details** (including positioning & motion)
- ▶ we can often not pin down the details (i.e., identify sub-structures or sub-pathways)
- ▶ we can usually not say how exactly the parts work together (especially cross-modal)

Neural processing of visual information -2-

- ▶ we cannot yet explain invariance & robustness, top-down influences, substantial individual differences, development & injury recovery
- ▶ it especially remains **unclear how the overall distributed representations are integrated** (e.g. by synchronization or convergence of neural connections), especially across modalities, if ambiguous or in conflict and for (exponentially) complex objects
- ▶ there is a decent amount of room for the assumptions made by model A and the hypothesis of human minds processing Platonic abstract objects

Neuroscience experiments / outlook

- ▶ **from quantum theory to cognition**; in A-world Neuroscience we could look at quantum theory for experiments; similarities are found because objects at the bottom are not yet individualized, those at the top not anymore
- ▶ 'Neuro double slit experiments'; does brain processing look as if intermediates are missing?
- ▶ 'Neuro Bell experiments'; do brain processing statistics indicate 'entanglement', i.e. simultaneous, distant activity related to same content
- ▶ 'Neuro Kochen-Specker experiments'; do brain processing statistics indicate 'contextuality', i.e. same activity delivers different contents?
- ▶ ...
- ▶ **and back from cognition to quantum theory**; if the link holds, can abstract object cognition serve as blue print for the development of quantum computing algorithms?
- ▶ ... but unlike abstract object manipulation, quantum computing can make use of only few and sharp (material) semantic implications

Appendix II: Evolutionary monad mechanics

Evolutionary monad mechanics – a toy model

Physical monad life

- ▶ initially monads acquire random properties, using other properties as 'bridges'; evolution is a mess, stable only after eons
- ▶ simple monads are not able to acquire ideas of proper physics or complex concepts like change, active/passive, etc.; only qualia-type properties like emotions etc.
- ▶ good/bad (is good the start of all growth?), then e.g. good+sweet, then it feels better to change sweet level if good+sweet+unwanted-import happens, etc.
- ▶ gives rise to 'taste position', i.e. functional space from 'taste spectrum'; no particle shall have the exact same taste
- ▶ universal passing of objects requires at least three dimensions (so sweet, salty, umami then?)
- ▶ we have a 'particle' (as abstract object) individuated in functional space now
- ▶ such a particle becomes physical, if all relevant properties are handled (overall) consistently by all relevant monads, and that in turns guarantees its identity

Evolutionary monad mechanics/a toy model -2-

The sweeter life virus

- ▶ 'bare' monad acquires 'sweet' and 'sweeter' properties
- ▶ rushes along the sweetness spectrum; can 'infect' our 'particles' and takes them along
- ▶ transfer of 'impulse' between 'particles' (stops X-1/starts anew at X on spectrum)
- ▶ the 'lower' level monad serves as a endosymbiont for the 'higher'-level monad, which sees additional higher-level properties
- ▶ the lower monad can invoke additional properties in its part of the higher-level object
- ▶ higher monads may at some point get rid of taste properties, but still see other 'signaling' properties (qualia) from lower monads
- ▶ higher monad 'particle' objects then move in functional space without the higher monads being aware of the mechanics anymore
- ▶ physical laws would be the outcome of the evolution of a monad population (somewhat like, but more flexible than Stephen Wolfram's New Kind of Science)
- ▶ this would require a biological reading of physics through and through; results from evolution, ecology etc. become relevant for Physics ...

Evolutionary monad mechanics / outlook

- ▶ we don't know about monad experience in the same way in which we do not know how echolocation feels – but that does not mean that it is impossible to map out underlying semantic implications or resulting structural relations
- ▶ semantical implications of echolocation qualia somehow similar to colors, shapes, etc.
- ▶ should be possible to formalize this and then simulate it
- ▶ maybe via a description similar to Boolean operations or computational gates
- ▶ when can then look to quantum theory for experiments
- ▶ e.g., $x/y/z$ properties should be independent; are there 'hidden **split** variables'?